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Multimodal Remote Sensing Image Classification via Hierarchical Frequency-Enhanced Mamba Fusion Network

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***Abstract*—Hyperspectral image (HSI) and light detection and ranging (LiDAR) data provide complementary characteristics that improve the performance of remote sensing image classification. However, existing multimodal fusion methods still struggle against two critical challenges. First, most cross-modal fusion strategies adopt overly simplistic feature interaction designs, which prevents them from fully exploiting the complementary information between the two modalities. Second, although the Mamba architecture efficiently models long-range dependencies, its inherent tendency toward excessive smoothing limits its ability to capture high-frequency details. To address these challenges, we propose a Hierarchical Frequency-Enhanced Mamba Fusion Network (HFMNet) for multimodal remote sensing image classification, which enhances global representation capability by integrating state-space modeling with frequency-enhanced analysis. Specifically, the Fourier transform is embedded into the Spatial Mamba modules to compensate for their limited sensitivity to high-frequency variations, thereby enhancing the representation of fine-grained spatial details while maintaining efficient long-range dependency modeling. Furthermore, a multi-level dual-attention-guided fusion module is developed to explicitly integrate multimodal features along both spatial and channel dimensions. This design facilitates the alignment and complementary exploitation of heterogeneous information. The level-specific fused representations are then progressively aggregated in a multi-level manner to strengthen multi-level feature consistency and improve classification reliability. Under a challenging few-shot learning scenario (only 20 training samples per class), comprehensive experiments on Houston2013, MUUFL, and Trento show that our method obtains OA values of 95.45%, 86.08%, and 99.35%, respectively. This demonstrates its robust feature extraction capabilities under sparse supervision, and it performs favorably against state-of-the-art approaches.**

***Index Terms*—Hyperspectral image (HSI); Light detection and ranging (LiDAR); Fourier transform; Mamba; Attention mechanism**

I. Introduction

With advances in remote sensing, hyperspectral images (HSI) and light detection and ranging (LiDAR) have become fundamental data sources for multimodal data fusion [1], [2]. HSI captures fine-grained spectral variations of surface materials, enabling discrimination of land-cover classes with highly similar spectral signatures. In contrast, LiDAR provides precise three-dimensional structural information, offering complementary geometric cues for terrain and scene analysis [3]. Owing to their spectral–structural complementarity, effectively integrating HSI and LiDAR to enhance discriminative representations remains a key challenge [4]–[6].

Current multimodal remote sensing methods are mainly based on CNNs, Transformers, and emerging sequence models such as Mamba [7]. Due to their local inductive bias, CNNs effectively extract local features such as edges and textures for land-cover classification. For instance, Peng *et al.* [8] propose the Multi-View Hierarchical Network that leverages band partitioning and hierarchical fusion to exploit heterogeneity across HSI subsets. Similarly, Zhang *et al.* [9] put forward the Patch-to-Patch Convolutional Neural Network, which allows for the collaborative modeling of complementary features derived from HSI and LiDAR data. However, complex remote sensing scenes require modeling long-range dependencies and large-scale spatial structures. Limited receptive fields make CNNs inadequate for capturing long-range contextual dependencies [10].

Transformers leverage self-attention to explicitly model global contextual dependencies beyond local convolutions. Li *et al.* [11] propose a Sliding Window Convolutional Transformer that integrates sliding-window attention with local convolutions, thereby preserving fine-grained spatial details while modeling long-range dependencies. However, increasing image resolution significantly amplifies the computational cost of Transformer-based models [12]. Mamba, a state-space model, shows clear advantages in long-sequence modeling. Liao *et al.* [13] introduce Mamba for HSI-LiDAR joint modeling and investigate its potential for integrating heterogeneous features. Li *et al.* [14] propose the Spatial–Spectral Mamba framework that effectively captures joint spectral–spatial features from high-dimensional data while maintaining low computational complexity. With linear-time complexity, Mamba is well suited for long sequences and high-resolution remote sensing tasks.

Nonetheless, existing Mamba-based methods often rely on one-dimensional (1D) sequence modeling or local convolutions, which compromises two-dimensional (2D) spatial topology and restricts global structure modeling [15]. Furthermore, prevailing frameworks typically process each modality independently and rely on simple aggregation strategies. This isolated processing lacks unified cross-modal interaction and underexploits the inherent complementarity of HSI and LiDAR data, thereby limiting the potential of inter-modal integration across various downstream tasks (e.g., land-cover classification

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[8] and cross-modal retrieval [16]). Moreover, unlike tailored feature enhancement models [17]–[19], many existing studies fail to adequately preserve modality-specific characteristics during fusion. This often introduces redundant or interfering information, degrading the discriminative power of fused features. Under complex remote sensing conditions, these fusion strategies struggle to learn stable and robust cross-modal features [20], [21], especially in complex regions with hard-to-classify patterns such as spatial boundaries and shadows [22]. Consequently, classification accuracy and generalization performance remain limited.

To address these limitations, we propose a Hierarchical Frequency-Enhanced Mamba Fusion Network (HFMNet) for multimodal remote sensing image classification. HFMNet integrates long-range dependency modeling with global spectral perception in the frequency domain. This integration enhances the model's global representation capacity across multi-level scenes. Specifically, we design a Hierarchical Frequency–Mamba Encoding (HFME) module. This module introduces a Fourier-based frequency-enhanced mechanism to compensate for the Mamba's limited ability to capture high-frequency details. To further alleviate insufficient cross-modal interactions and severe feature redundancy, a Multi-Level Dual-Attention-Guided Fusion (ML-DAGF) module is developed. Finally, a multi-level feature aggregation module progressively integrates the fused representations across levels, thereby enhancing discriminative capability for diverse land-cover categories.

The main contributions of this work are summarized as:

1) We propose a HFME module that integrates a Fourier-based global frequency modeling and high-frequency extraction into the Mamba architecture. This design directly compensates for Mamba's inherent insensitivity to high-frequency components, strengthening the network's ability to capture and preserve fine-grained structural details for more discriminative spatial representations.
2) A ML-DAGF module is designed to synergize heterogeneous remote sensing data. By jointly exploiting spatial and channel attention across multiple semantic levels, it effectively extracts complementary cross-modal features while strictly preserving crucial modality-specific characteristics, thereby boosting overall classification performance.
3) Ten-fold independent experiments on three popular benchmark datasets verify that, even under a challenging few-shot learning scenario (only 20 samples per class), the proposed approach consistently yields better performance than state-of-the-art counterparts, demonstrating its robust feature extraction capabilities under sparse supervision.

The remainder of this paper is organized as follows. In Section II, we review mainstream modeling frameworks for multimodal remote sensing classification and representative multimodal fusion methods. In Section III, the architecture and structural details of HFMNet are presented. Section IV compares HFMNet with SOTA methods and conducts comprehensive ablation studies to evaluate its performance and effectiveness. Finally, Section V summarizes the main contributions and findings of this paper.

II. Related works

A. Multimodal Classification Architectures

In multimodal remote sensing classification, deep learning architectures have been widely explored to integrate heterogeneous modalities and enhance feature representation. CNN-based methods effectively capture local spatial features, motivating representative works such as the Interleaving Perception CNN [23] and HypsLiDNet [24], which improve multimodal fusion through architectural design and attention mechanisms. However, their limited ability to model long-range dependencies constrains performance in complex scenes. To alleviate this limitation, Transformer-based approaches leverage self-attention to capture global context, as demonstrated by cross-source Transformers [25] and cross-attention-guided networks [26]. Despite their effectiveness, the high computational cost and training complexity of Transformers remain major bottlenecks for large-scale remote sensing applications.

To address the efficiency limitations of Transformers, state-space modeling approaches based on Mamba have recently attracted increasing attention. Zhang *et al.* [27] develop CenterMamba, which employs a center-scanning mechanism to enhance semantic representations of key regions. Similarly, Liu *et al.* [28] design HyperMamba with adaptive modules to optimize performance on high-dimensional HSI data. Although these Mamba-based architectures enable efficient sequence modeling, they still struggle with restricted 2D spatial structure modeling and suboptimal cross-modal alignment.

To overcome this spatial limitation and effectively preserve fine-grained structures, we introduce the HFME module. By leveraging Fourier-based frequency extraction, this design specifically repairs Mamba's vulnerability to high-frequency information loss, establishing a structurally rich feature foundation for subsequent cross-modal integration.

B. Multimodal Fusion Methods

In multimodal remote sensing classification, effectively fusing heterogeneous data across modalities remains a major bottleneck. To enhance the discriminability of modality-specific features, attention mechanisms are widely adopted. For example, Cai *et al.* [29] explicitly modeled inter-node relationships via graph attention, while Fan *et al.* [30] and Ge *et al.* [31] focused on multi-scale learning and local cross-attention to emphasize regional responses. Although these methods partially alleviate information dilution, most still focus on intra-modal modeling and lack semantic-level collaborative mechanisms. To facilitate deeper semantic alignment, subsequent studies have shifted toward cross-modal interaction fusion during the encoding stage. Wang *et al.* [32] and Cao *et al.* [33] constructed deep semantic pathways using Transformer cascades and spectral–spatial–language trimodal frameworks, respectively, while Tu *et al.* [34] utilized feature graph matching to guide the joint feature space. Despite these diverse designs, existing fusion methods share two common limitations: 1) the lack of synergistic spatial-channel pathways across multiple semantic levels; and 2) a tendency toward information homogenization, where deep cross-modal

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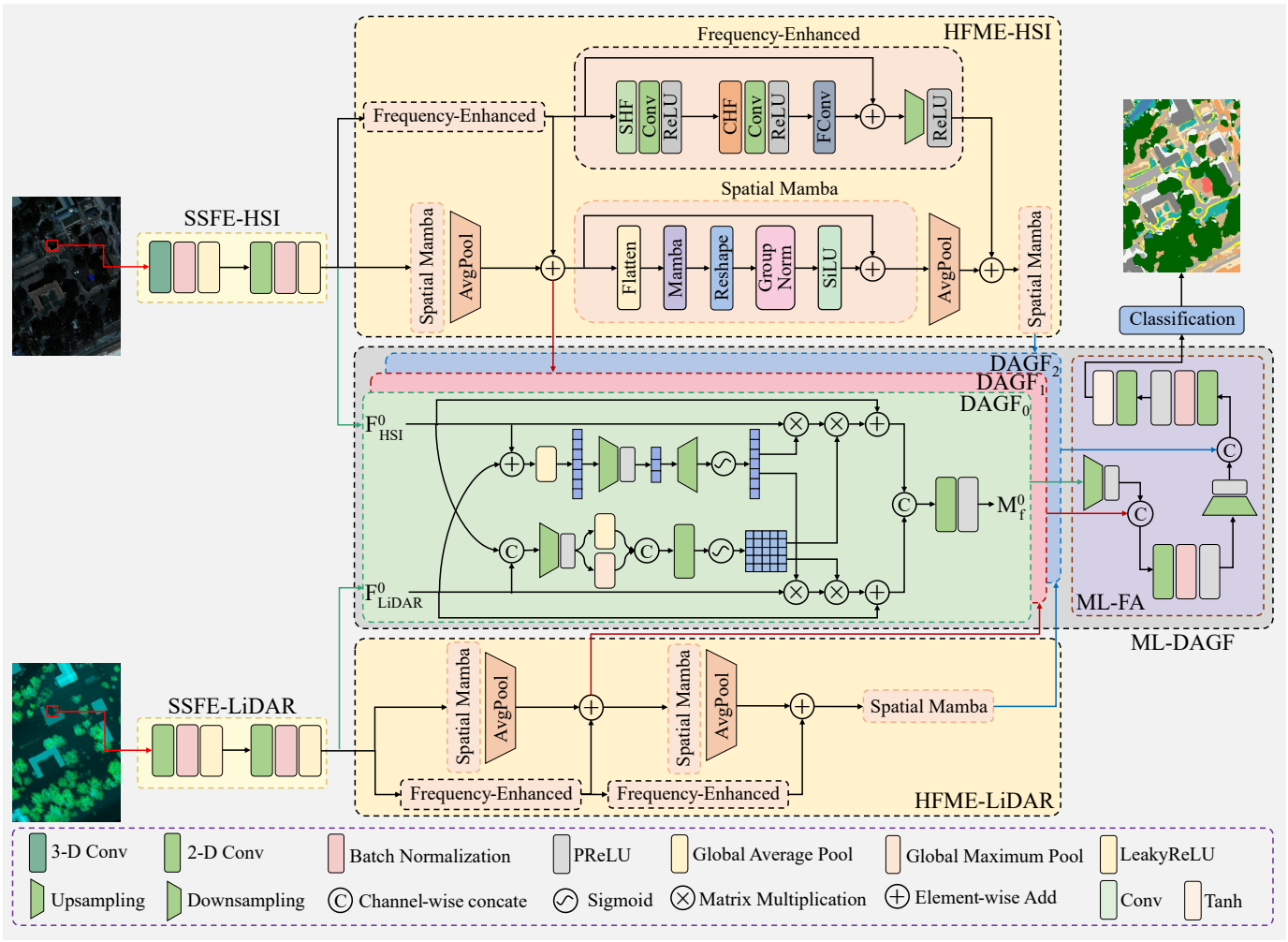


Fig. 1: Overall architecture of the HFMNet method. HSI and LiDAR data are first processed by the Spatial–Spectral Feature Extraction (SSFE) module to obtain spatial and channel-wise features. These features are then encoded by the Hierarchical Frequency–Mamba Encoding (HFME) module, which integrates Spatial Mamba for long-range dependency modeling and Frequency-Enhanced modules for high-frequency feature extraction and global frequency modeling. Subsequently, the Multi-Level Dual-Attention-Guided Fusion (ML-DAGF) module performs multi-level feature fusion along spatial and channel dimensions via Dual-Attention-Guided Fusion (DAGF) and Multi-Level Feature Aggregation (ML-FA). Finally, a classifier outputs the predicted labels. SHF: Spatial High-Frequency; CHF: Channel High-Frequency; Fconv: Frequency Domain Convolution.

interactions inadvertently suppress valuable modality-specific characteristics.

To explicitly address these bottlenecks, our proposed ML-DAGF module is designed to balance cross-modal integration with modality-specific preservation. By embedding comprehensive spatial-channel pathways within a multi-level framework, it ensures the effective extraction of deep complementary semantics without sacrificing the unique attributes of individual modalities.

III. Proposed Method

This work introduces a hierarchical frequency-enhanced Mamba fusion network, termed HFMNet, for multimodal remote sensing image classification. Illustrated in Fig. 1, our network comprises four components: Spatial–Spectral Feature Extraction (SSFE), Hierarchical Frequency–Mamba Encoding (HFME), Multi-Level Dual-Attention-Guided Fusion (ML-DAGF), and the classification and loss. This section elaborates on the design details of each component.

A. Spatial-Spectral Feature Extraction Module

The SSFE module adopts a dual-branch architecture, in which each branch independently processes one modality and consists of two neural network layers. Each convolutional layer consists of a convolution operation, followed by a batch normalization layer and a LeakyReLU activation function. The image patches fed into each branch are extracted from HSI and LiDAR data, respectively, yielding $X_{HSI} \in \mathbb{R}^{b\times c_1\times p\times p}$ and $X_{LiDAR} \in \mathbb{R}^{b\times c_2\times p\times p}$. Here, b denotes the batch size, $p \times p$ represents the patch size while c_1 and c_2 correspond to spectral band count in HSI and channel count in LiDAR data, respectively.

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1) LiDAR Branch: The LiDAR branch adopts two 1×1 convolutional layers with 32 and 64 kernels, respectively, to extract elevation-related features.

2) HSI Branch: The first convolutional layer in the HSI branch is a three-dimensional (3D) convolution with a kernel size of $5 \times 1 \times 1$. Specifically, the input HSI cube X_{HSI} is first expanded by inserting an additional channel dimension using an unsqueeze operation, forming a tensor of size $(b, 1, c_1, p, p)$ that serves as the input to the 3D convolution. The 3D convolution employs 8 output channels to efficiently extract spectral–spatial features from HSI data. The resulting features are then reshaped using a rearrangement operation to restore the original spatial structure and subsequently processed by 64 1×1 convolutional kernels for further feature transformation.

After this step, the patches from the two modalities are mapped to $F^0_{LiDAR} \in \mathbb{R}^{b \times c' \times p \times p}$ and $F^0_{HSI} \in \mathbb{R}^{b \times c' \times p \times p}$, where $c' = 64$.

B. Hierarchical Frequency–Mamba Encoding

Multimodal remote sensing image classification is formulated as a pixel-level classification task, where each pixel is represented by a discriminative feature embedding. Recently, Mamba is adopted in this area owing to its linear-time state-space formulation, which enables efficient modeling of long-range dependencies in spectral and elevation sequences. However, Mamba-based modeling is inherently influenced by the ordering of flattened tokens, which may limit its ability to jointly capture global and local spatial patterns.

The HFME module leverages Spatial Mamba to extract multi-level features from the input images. Simultaneously, the Fourier-based Frequency-Enhanced module is applied to map the images into the frequency domain, capturing explicit high-frequency structural details and enabling global frequency modeling.

Regarding the multimodal inputs, although HSI and LiDAR exhibit inherent heterogeneity, they are first projected into a unified, dimensionally consistent semantic space ($c' = 64$ channels) by the initial modality-specific SSFE module. Consequently, applying an identical HFME architecture to both branches offers two critical advantages:1) it ensures the symmetric extraction of long-range dependencies without introducing architecture-induced bias; and 2) it guarantees identical receptive fields and semantic abstraction levels, which is vital for robust cross-modal alignment in the subsequent fusion stage. Given this symmetric design, the following formulation is presented using the HSI branch as an example.

1) Spatial Mamba: Spatial Mamba is adopted from [14]. The forward process can be formulated as follows:

$$MF^j = \mathrm{Reshape}\big(\mathrm{Mamba}\big(\mathrm{Flatten}(X^j)\big)\big), \tag{1}$$

$$F^j_{SM} = \mathrm{SiLU}\big(\mathrm{GN}\big(MF^j\big)\big) + X^j, \tag{2}$$

where $X^j \in \mathbb{R}^{b \times c' \times p_j \times p_j}$ denotes the input feature map of the j-th Spatial Mamba module, and F^j_{SM} denotes the corresponding output feature, with $j \in \{0, 1, 2\}$. Here, $p_j = \frac{p}{2^j}$ represents the dynamically reduced spatial resolution at level j. To accommodate the 1D sequence processing requirement of Mamba, the $\mathrm{Flatten}(\cdot)$ operation linearly unfolds the spatial dimensions of X^j into a 1D sequence of length $p_j \times p_j$ while strictly preserving the channel depth c'. Following the state-space modeling, the $\mathrm{Reshape}(\cdot)$ deterministically maps this sequence back to its original 2D spatial topology $\mathbb{R}^{b \times c' \times p_j \times p_j}$. This structural restoration spatially re-aligns the Mamba-derived long-range dependencies with the extracted high-frequency details and global frequency representations, enabling seamless element-wise addition for a comprehensive joint representation. Here, $\mathrm{GN}(\cdot)$ is group normalization, and $\mathrm{SiLU}(\cdot)$ denotes the Sigmoid Linear Unit activation.

2) Frequency-Enhanced: Due to the limited capability of Spatial Mamba in global image modeling, high-frequency information tends to be lost as network depth increases. Such information is critical for accurate multimodal remote sensing image fusion classification. The Frequency-Enhanced module maps features into the Fourier domain to explicitly capture high-frequency information and refine it, thereby enhancing the network's global modeling capability.

High-frequency feature extraction: The spatial high-frequency (SHF) module and the channel high-frequency (CHF) module convert feature maps into the frequency domain via the fast Fourier transform (FFT) and extract high-frequency components using dedicated high-frequency filters. To accurately capture high-frequency details, the filters must be dynamically adapted to the dimensional properties of the input tensor. Let $X_{in} \in \mathbb{R}^{b \times c_{in} \times h_{in} \times w_{in}}$ denote the generic input feature map. It is crucial to note that m, n, and u are zero-centered frequency indices obtained after the $\mathrm{fftshift}(\cdot)$ operation. Specifically, the SHF filter W_S operates on the 2D spatial frequency domain. With the cutoff ratio set empirically to 4, it is defined as:

$$W_S(f_{m,n}) = \begin{cases} 0 & ,\text{if } |m| < \frac{h_{in}}{4} \text{ and } |n| < \frac{w_{in}}{4}, \\ f_{m,n} & ,\text{otherwise}, \end{cases} \tag{3}$$

where $f_{m,n}$ denotes the spectral component at spatial frequency position (m, n). Conversely, the CHF filter W_C is applied strictly along the 1D channel dimension, defined as:

$$W_C(f_u) = \begin{cases} 0 & ,\text{if } |u| < \frac{c_{in}}{4}, \\ f_u & ,\text{otherwise}, \end{cases} \tag{4}$$

where f_u represents the spectral component at the 1D channel frequency index u. The filtered frequency-domain representation f_{dim} is formulated as:

$$f_{dim} = W_{dim}(\mathrm{fftshift}(\mathcal{F}_{dim}(X_{in}))),\ dim \in \{S, C\}, \tag{5}$$

where $\mathcal{F}_{dim}(\cdot)$ denotes the FFT applied along a specific dimension (2D for S and 1D for C), and $\mathrm{fftshift}(\cdot)$ shifts the zero-frequency component to the center of the spectrum.

After filtering, an inverse shift operation $\mathrm{ifftshift}(\cdot)$ is performed prior to the IFFT to map the features back to the spatial domain. The real part is extracted and passed through a Rectified Linear Unit (ReLU) activation function to obtain the enhanced representation X'_{dim}:

$$X'_{dim} = \mathrm{ReLU}(\mathrm{Re}(\mathcal{F}^{-1}_{dim}(\mathrm{ifftshift}(f_{dim})))), dim \in \{S, C\}, \tag{6}$$

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where $\mathcal{F}_{dim}^{-1}(\cdot)$ denotes the IFFT applied along the specific dimension. The real-part operator $\mathrm{Re}(\cdot)$ is applied solely to eliminate negligible imaginary residues caused by numerical precision. This overall process enhances the network's ability to preserve fine-grained structural details that are essential for accurate multimodal remote sensing image classification.

For brevity, this entire sequence is encapsulated as $X'_S = \mathrm{SHF}(X_{in})$ and $X'_C = \mathrm{CHF}(X_{in})$ for the spatial and channel dimensions, respectively.

Frequency Domain Convolution: To enhance global modeling capability, Frequency Domain Convolution projects the input feature map X_{in} into the frequency domain via FFT $\mathcal{F}(\cdot)$ and centers it using $\mathrm{fftshift}(\cdot)$. The real-part operator $\mathrm{Re}(\cdot)$ and imaginary-part operator $\mathrm{Im}(\cdot)$ are then applied to obtain the real and imaginary components of the frequency-domain representation. These components are refined independently using dedicated convolution operators $\mathcal{C}_{\mathrm{real}}(\cdot)$ and $\mathcal{C}_{\mathrm{imag}}(\cdot)$, respectively:

$$F'_{\mathrm{Re}} = \mathcal{C}_{\mathrm{real}}\left(\mathrm{Re}\left(\mathrm{ifftshift}(\mathcal{F}(X_{in}))\right)\right), \tag{7}$$

$$F'_{\mathrm{Im}} = \mathcal{C}_{\mathrm{imag}}\left(\mathrm{Im}\left(\mathrm{ifftshift}(\mathcal{F}(X_{in}))\right)\right), \tag{8}$$

where F'_{Re} and F'_{Im} denote the refined real and imaginary features after convolution, respectively.

The refined components are recombined into a complex-valued tensor F':

$$F' = F'_{\mathrm{Re}} + i \cdot F'_{\mathrm{Im}}, \tag{9}$$

where the imaginary unit i satisfies $i^2 = -1$. Because independent convolutions alter the Hermitian symmetry, the IFFT inherently yields a complex-valued tensor. To map this signal back to the spatial domain and ensure architectural compatibility with subsequent real-valued neural network layers, we apply $\mathrm{ifftshift}(\cdot)$, perform the IFFT $\mathcal{F}^{-1}(\cdot)$, and explicitly extract its real component. Then, it is processed through a ReLU activation function, which injects non-linearity to produce enhanced global features X_{out} without altering the original spatial resolution:

$$X_{out} = \mathrm{ReLU}\big(\mathrm{Re}\big(\mathcal{F}^{-1}\big(\mathrm{ifftshift}\big(F'\big)\big)\big)\big). \tag{10}$$

To account for the architectural difference in block counts, we denote the input feature for the k-th Frequency-Enhanced module specifically as Y^k ($k \in \{0, 1\}$). The recursive relationship is explicitly defined with the initial input $Y^0 = F^0_{HSI}$ and the subsequent input updated as $Y^1 = F^0_{Freq}$. The sequential propagation is:

$$TF^k_S = \mathrm{ReLU}\left(\mathrm{Conv}\left(\mathrm{SHF}(Y^k)\right)\right), \tag{11}$$

$$TF^k_C = \mathrm{ReLU}\left(\mathrm{Conv}\left(\mathrm{CHF}(TF^k_S)\right)\right), \tag{12}$$

$$F^k_{Freq} = \mathrm{ReLU}\left(\mathrm{Conv}_{S\downarrow}\left(\mathrm{FConv}(TF^k_C) + Y^k\right)\right), \tag{13}$$

where $\mathrm{FConv}(\cdot)$ denotes the frequency domain convolution, and $\mathrm{Conv}(\cdot)$ is a convolutional layer. TF^k_S denotes the refined high-frequency spatial representation obtained after convolutional mapping, and TF^k_C denotes the subsequent refined representation of high-frequency variations along the learned channel dimension. $\mathrm{Conv}_{S\downarrow}(\cdot)$ denotes a spatial downsampling convolution that aligns the feature resolution with the subsequent Spatial Mamba input for cross-branch addition. F^k_{Freq} denotes the discriminative high-frequency representation learned by the Frequency-Enhanced module. By selectively emphasizing informative high-frequency components while suppressing spurious frequency responses, this module enhances feature robustness and reduces sensitivity to high-frequency noise present in the original data.

Multi-Level Feature Aggregation in HFME: Finally, to construct the hierarchical representations, the outputs from the parallel branches are aggregated. Specifically, X^0 corresponds to the output feature map F^0_{HSI} from the SSFE module. For the subsequent levels j, the input X^j is calculated by element-wise addition of the pooled Spatial Mamba feature and the frequency-enhanced feature from the preceding level:

$$X^j = \mathrm{AvgPool}(F^{j-1}_{SM}) + F^{j-1}_{Freq}, \quad j \in \{1, 2\}, \tag{14}$$

where $\mathrm{AvgPool}(\cdot)$ denotes an average pooling operation with a receptive field size and stride of 2. By progressively aggregating these branches, the HFME module produces multi-level feature maps with spatial resolutions of $p \times p$, $\frac{p}{2} \times \frac{p}{2}$ and $\frac{p}{4} \times \frac{p}{4}$. These feature maps primarily capture long-range contextual dependencies and broad spatial layouts, representing modality-specific features across multiple semantic levels.

C. Multi-Level Dual-Attention-Guided Fusion

Despite the integration of attention mechanisms in HSI–LiDAR fusion, existing methods are usually confined to single-modality spatial or channel feature refinement and lack a unified cross-modal interaction paradigm. This deficiency leads to the underexploitation of inherent complementarity and intrinsic correlations between HSI and LiDAR modalities, thereby degrading the discriminative power of the fused representations. To address this issue, we propose a ML-DAGF module, which effectively exploits multi-level semantic complementarity between HSI and LiDAR and consists of a Dual-Attention-Guided Fusion (DAGF) module and a Multi-Level Feature Aggregation (ML-FA) module.

1) Dual-Attention-Guided Fusion: The DAGF module consists of a spatial-attention branch and a channel-attention branch, which are designed to suppress redundant information along spatial and channel dimensions while emphasizing task-relevant cues. The resulting attention maps are applied to the modality-specific features to selectively enhance shared informative components while preserving unique structural characteristics.

In this paper, DAGF module takes as input multi-level feature maps (three levels per modality) produced by the HFME module. For clarity, the same resolution feature maps, F^j_{HSI} and F^j_{LiDAR}, are used as illustrative examples.

Channel Attention Module: To establish an explicit cross-modal correlation between HSI spectral bands and LiDAR elevation signatures, complementary information from the two branches is first aggregated via element-wise addition. Subsequent global average pooling compresses the 2D spatial features into a 1D channel representation $CG^j \in \mathbb{R}^{b \times c' \times 1 \times 1}$, capturing a global semantic consensus. The channel dimension is first compressed using a 1×1 convolution and then restored

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via convolution-based channel upsampling. As a result, a channel attention map CM^j is obtained as follows:

$$CG^j = \mathrm{GAP}\big(F^j_{HSI} + F^j_{LiDAR}\big), \tag{15}$$

$$CM^j = \sigma\big(\mathrm{Conv}_{C\uparrow}\big(\mathrm{PReLU}\big(\mathrm{Conv}_{C\downarrow}\big(CG^j\big)\big)\big)\big), \tag{16}$$

where F^j_{HSI} and F^j_{LiDAR} represent the feature maps at the j-th level, $\mathrm{GAP}(\cdot)$ denotes global average pooling; $\mathrm{Conv}_{C\downarrow}(\cdot)$ denotes a channel reduction convolution that halves the channel dimension, whereas $\mathrm{Conv}_{C\uparrow}(\cdot)$ denotes a channel expansion convolution that restores it. $\mathrm{PReLU}(\cdot)$ and $\sigma(\cdot)$ denote the Parametric Rectified Linear Unit and the Sigmoid activation function, respectively.

Spatial Attention Module: To precisely locate spatial boundaries and morphological edges, dual-modality feature maps sharing the same spatial resolution are concatenated along channels and processed via convolution with activation, yielding SG^j. Next, both channel average pooling and channel max pooling operate on SG^j along the channel dimension to aggregate geometric contexts. Their channel-concatenated results pass through a convolution layer to produce a shared cross-modal spatial attention map SM^j, utilizing the aggregated geometric contexts to mutually enhance both modalities, expressed as:

$$SG^j = \mathrm{PReLU}\big(\mathrm{Conv}_{C\downarrow}\big(\mathrm{Cat}(F^j_{HSI},\, F^j_{LiDAR})\big)\big), \tag{17}$$

$$SM^j = \sigma\big(\mathrm{Conv}\big(\mathrm{Cat}\big(\mathrm{CAP}\big(SG^j\big),\, \mathrm{CMP}\big(SG^j\big)\big)\big)\big), \tag{18}$$

where $CAP(\cdot)$ and $CMP(\cdot)$ denote average pooling and max pooling operations applied along the channel dimension, respectively. $\mathrm{Cat}(\cdot)$ denotes channel-wise concatenation. Through these steps, 1D channel attention and 2D spatial attention are obtained and subsequently used for multimodal deep feature fusion.

$$M^j_{HSI} = SM^j \otimes (CM^j \otimes F^j_{HSI}) + F^j_{HSI}, \tag{19}$$

$$M^j_{LiDAR} = SM^j \otimes (CM^j \otimes F^j_{LiDAR}) + F^j_{LiDAR}, \tag{20}$$

$$M^j_f = \mathrm{PReLU}\big(\mathrm{Conv}\big(\mathrm{Cat}\big(M^j_{HSI},\, M^j_{LiDAR}\big)\big)\big), \tag{21}$$

where M^j_{HSI} and M^j_{LiDAR} denote the modality-specific feature maps after mutual guidance, and M^j_f denotes the corresponding fused feature map. Driven by a sequential dual-gating mechanism, the DAGF module selectively enhances informative spatial-channel responses while retaining the original features through residual connections. This design provides a direct residual path for retaining modality-specific information, effectively preventing information homogenization. Here, $\otimes$ denotes element-wise multiplication. Finally, fused feature representations M^0_f, M^1_f and M^2_f are obtained at three different spatial resolutions.

2) Multi-Level Feature Aggregation: To aggregate level-specific features (M^0_f, M^1_f, M^2_f) without the severe semantic aliasing caused by abrupt resolution discrepancies in direct concatenation, the ML-FA module employs an iterative, layer-by-layer fusion paradigm. By continuously refining intermediate representations via sequential downsampling and cascading, this progressive coupling provides superior non-linear semantic mapping while effectively propagating fine-grained structural details from the shallowest layer into the final representation. Finally, a Tanh activation function restricts the output's dynamic range to stabilize cross-level semantic consistency.

Specifically, to initialize the progressive fusion, we denote the base fuse feature as $TF_0 = M^0_f$. At each subsequent step $a \in \{1, 2\}$, the fused feature from the preceding level, TF_{a-1}, is first downsampled via a convolutional layer to match the spatial resolution of the current level feature M^a_f. The downsampled intermediate feature TM_a and M^a_f are then concatenated and fused using a subsequent convolutional layer to produce the updated fused representation TF_a. Finally, a convolutional layer with a hyperbolic tangent (Tanh) activation function is applied to TF_2 to generate the final multimodal and multi-level fusion output Out. The overall fusion process is summarized as follows:

$$TF_0 = M^0_f, \tag{22}$$

$$TM_a = \mathrm{PReLU}\left(\mathrm{Conv}_{S_down}\left(TF_{a-1}\right)\right), \tag{23}$$

$$TF_a = \mathrm{PReLU}\left(\mathrm{BN}\left(\mathrm{Conv}\left(\mathrm{Cat}\left(TM_a, M^a_f\right)\right)\right)\right), \tag{24}$$

$$Out = \mathrm{Tanh}\big(\mathrm{Conv}\big(TF_2\big)\big), \tag{25}$$

where $\mathrm{BN}(\cdot)$ denotes a batch normalization operation.

D. Classification and Loss Function

Finally, the fused HSI and LiDAR features are fed into a two-layer fully connected classifier. The classifier consists of a 128-dimensional hidden layer with LeakyReLU activation and dropout regularization, followed by an output layer that produces raw logits over C classes. These logits are then converted into predicted probabilities via a Softmax function. Training minimizes the cross-entropy loss to reduce the divergence between the predicted probability distribution and the ground-truth labels. For a single training sample, the loss is computed as:

$$\mathcal{L} = -\sum_{c=1}^{C} y_c \log(p_c),\ y_c \in \{0,\, 1\}, \tag{26}$$

where p_c denotes the predicted probability that the sample belongs to class c, and C is the total number of classes. Since y_c follows a one-hot encoding scheme, only the term corresponding to the ground-truth class contributes to Equation (26).

IV. Experiments

This section provides an overview of experimental datasets, evaluation metrics, and implementation settings. Comprehensive experiments are performed to substantiate the efficacy and advantages of HFMNet.

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A. Dataset Description

This section validates the proposed HFMNet on three widely used benchmark datasets containing paired HSI and LiDAR data.

1) Houston2013 Dataset [35]: Collected in June 2012 over the University of Houston campus and adjacent urban regions, the Houston2013 dataset provides co-registered HSI and LiDAR data at $349 \times 1,905$ pixels with a spatial resolution of 2.5 m. Its HSI component is captured by the CASI-1500 sensor with 144 spectral bands spanning 380–$1,050$ nm, accompanied by a single-band LiDAR elevation map. It contains $15,029$ annotated pixels across 15 land-cover categories, from which 20 samples per class are selected for training following [3].

2) MUUFL Dataset [36]: The MUUFL dataset is collected on November 8, 2010, over the University of Southern Mississippi campus, providing aligned HSI–LiDAR imagery at a spatial size of 325×220 pixels. The HSI data are obtained using the CASI-1500 sensor with 64 spectral bands covering 0.38–1.05 μm. The dataset includes $53,687$ labeled pixels from 11 categories, with 20 samples per class reserved for training as in [3].

3) Trento Dataset [37]: Acquired on July 25, 2011, in a rural region south of Trento, Italy, the Trento dataset offers paired HSI and LiDAR data with a spatial size of 600×166 pixels. Its HSI data are collected by the AISA Eagle sensor with 63 spectral bands ranging from 0.40 to 0.99 μm. It comprises $30,414$ labeled pixels grouped into six classes, with 20 randomly chosen samples per class for training.

B. Experimental Setting

All methods, including the proposed model, are evaluated on a system running Ubuntu 18.04 with an Intel CPU (2.60 GHz) and an NVIDIA A800 Tensor Core GPU (80 GB memory). Our model is developed in PyTorch using the Adam optimizer, with all parameters randomly initialized (no pretrained weights). The training epochs and batch size are set to 500 and 64, respectively, with the learning rate configured as 0.0002. To strictly evaluate the models under a challenging few-shot learning scenario, we employ a sparse supervision setting utilizing only 20 samples per class for training. To ensure a fair comparison, identical training and test splits are used for all datasets, with no overlap and without any data augmentation. Classification performance is quantitatively assessed using four metrics: overall accuracy (OA), average accuracy (AA), per-class accuracy, and the Kappa coefficient. All reported metrics are averaged over ten independent runs, with higher values indicating better performance.

C. Parameter Discussion

This section investigates the influence of patch size p on the comprehensive classification performance, specifically evaluating the OA, AA, and Kappa. It is widely acknowledged that image patches can provide valuable spatial contextual information for classification tasks, given that adjacent pixels within a local region typically belong to the same category. Consequently, the selection of an optimal patch size is of great significance for classification methods based on image patch input. The classification performance across various patch size settings is systematically summarized in Fig. 2.

D. Performance Comparison

We evaluate the efficacy of our proposed model through comparative experiments against several representative HSI-LiDAR fusion methods. These comparison methods include: Coupled-CNNs [38] proposes a feature-level and decision-level fusion model based on coupled convolutional networks; GLT-Net [39] combines CNN and Transformer architectures; Fusion_HCT [40] and HFSNet [41] focus on multimodal feature extraction and fusion; DSHFNet [42] incorporates dynamic multi-level attention mechanisms; AMSSE-Net [43] employs adaptive feature fusion strategies; MICF-Net [44] utilizes cross-modal attention mechanisms; MGMNet [45] adopts interactive guidance for multimodal feature extraction and fusion; DCU-Net [46] proposes a dynamic common-unique feature fusion network with cross dynamic-agent-attention; and MSAF [47] designs a masked self-attention fusion framework with cascaded cross-attention. All methods are reproduced using their official implementations and original hyperparameter settings.

1) Results on Houston2013 Dataset: The quantitative results on the Houston2013 dataset are summarized in Table I. HFMNet achieves the best performance across all three evaluation metrics among all the compared state-of-the-art methods in this study. Specifically, HFMNet improves OA and the Kappa coefficient over GLT-Net by 0.63 and 0.68 percentage points, respectively, and surpasses Fusion_HCT in AA by 0.19 percentage points. In terms of per-class accuracy, HFMNet ranks first or second in 6 of the 15 categories, indicating strong class-wise performance. Although not achieving the highest accuracy in every single category, HFMNet demonstrates highly competitive performance that is closely on par with the best-performing comparison methods. For instance, in the "Residential" category, HFMNet achieves an accuracy of 95.31%, trailing the top result (95.70%) by a mere 0.39 percentage points. Overall, classification accuracy across categories consistently exceeds 90%, indicating stable and reliable performance. The corresponding classification maps are presented in Fig. 3, along with those produced by other methods for visual comparison. As shown, the classification map produced by HFMNet is visually closest to the ground truth, further demonstrating the effectiveness of the proposed approach.

2) Results on MUUFL Dataset: On the MUUFL dataset (II), HFMNet consistently yields the highest classification performance among all the evaluated state-of-the-art methods, reaffirming the trend observed on Houston2013 data. In terms of OA, HFMNet outperforms competing approaches by margins ranging from 1.28 percentage points (GLT-Net) to 8.45 percentage points (AMSSE-Net). Per-class accuracy analysis further shows that the proposed model ranks first or second in 6 out of 11 land-cover categories. As shown in Fig. 4, the pixel-level classification maps from HFMNet exhibit the closest alignment with ground-truth labels. This superiority stems from effective multi-level exploitation of complementary multimodal information, yielding more discriminative feature representations.

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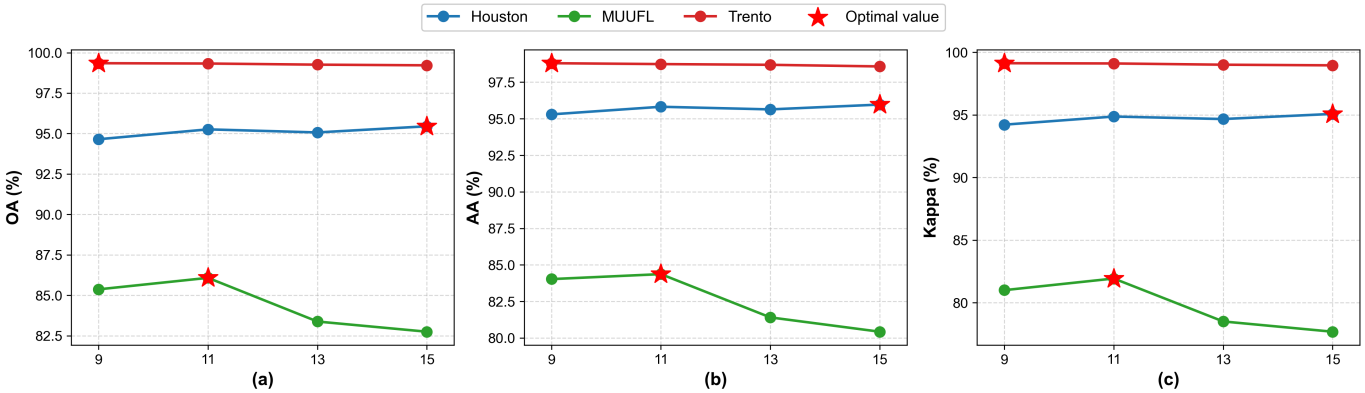


Fig. 2: Parameter analysis of p on the proposed HFMNet across different datasets. The subplots illustrate the variations in (a) OA, (b) AA, and (c) Kappa.

TABLE I: Classification performance of different methods on Houston2013 dataset (the best results are shown in **bold** and the second best ones are underlined)

No.	Class	Coupled-CNNs	GLT-Net	AMSSE-Net	Fusion_HCT	DSHFNet	HFSNet	MICF-Net	MGMNet	DCU-Net	MSAF	HFMNet
1	Healthy Grass	94.74	92.48	94.83	**98.62**	91.45	92.26	89.51	90.71	92.49	92.17	93.50
2	Stressed Grass	96.05	96.79	97.04	**97.57**	96.20	93.19	93.74	96.22	96.38	93.78	96.63
3	Synthetic Grass	99.68	99.39	99.76	99.85	**99.93**	**99.93**	99.25	99.79	99.79	99.76	99.68
4	Trees	98.42	97.47	97.69	98.28	97.53	97.17	95.82	96.18	98.85	95.65	**99.18**
5	Soil	99.64	99.82	99.24	**100.00**	99.71	99.20	98.92	99.30	98.94	98.99	99.82
6	Water	99.05	96.62	94.07	**100.00**	95.74	93.02	95.02	96.79	96.07	90.03	96.16
7	Residential	95.00	**95.70**	90.85	95.03	85.54	92.13	95.63	94.58	95.09	91.44	95.31
8	Commercial	89.86	88.52	82.47	**93.30**	86.73	85.36	84.09	86.92	85.20	82.57	91.26
9	Road	85.34	**94.26**	70.28	77.35	62.51	84.32	89.97	88.27	85.07	80.93	93.84
10	Highway	93.45	91.68	90.45	**94.12**	90.21	81.70	88.00	87.23	88.74	76.98	92.28
11	Railway	95.19	**95.48**	90.66	88.81	89.61	92.72	92.40	94.14	93.33	88.83	95.43
12	Parking Lot 1	88.12	87.22	85.49	**93.73**	91.91	88.54	85.49	89.84	90.96	84.62	91.76
13	Parking Lot 2	98.62	98.55	89.06	**100.00**	93.96	91.22	97.88	94.83	97.73	92.49	96.55
14	Tennis Court	**100.00**	99.95	99.46	**100.00**	99.78	99.68	99.98	99.73	99.88	98.90	**100.00**
15	Running Track	99.66	99.84	98.77	**100.00**	99.86	99.31	99.97	99.77	**100.00**	98.97	99.20
	OA(%)	94.57	94.82	91.06	94.74	90.66	91.79	92.64	93.39	93.61	90.02	**95.45**
	AA(%)	95.52	95.59	92.01	95.78	92.05	92.65	93.74	94.29	94.57	91.08	**95.97**
	Kappa(%)	94.13	94.40	90.33	94.31	89.90	91.12	92.04	92.85	93.09	89.21	**95.08**

TABLE II: Classification performance of different methods on MUUFL dataset (the best results are shown in **bold**, and the second best ones are underlined)

No.	Class	Coupled-CNNs	GLT-Net	AMSSE-Net	Fusion_HCT	DSHFNet	HFSNet	MICF-Net	MGMNet	DCU-Net	MSAF	HFMNet
1	Trees	88.33	90.17	83.00	85.41	79.90	87.45	88.97	86.92	89.41	85.22	**91.13**
2	Mostly Grass	75.05	76.71	73.33	77.54	**81.10**	74.70	73.52	72.52	73.58	74.64	75.93
3	Mixed Ground Surface	59.99	67.98	55.11	58.62	57.99	63.55	65.17	57.83	61.97	61.15	**69.66**
4	Dirt and Sand	86.62	83.90	79.03	87.82	**93.03**	84.53	84.96	75.53	80.05	82.35	86.85
5	Road	79.19	83.19	72.06	71.15	**90.64**	84.65	82.03	78.85	80.64	82.43	86.93
6	Water	99.19	99.28	98.12	99.08	**99.82**	98.34	99.01	98.83	98.61	98.00	99.51
7	Building Shadow	84.70	88.64	83.28	83.61	**92.32**	89.38	86.91	82.93	88.49	84.95	87.84
8	Building	93.87	93.64	92.28	91.11	90.78	93.38	92.74	92.46	93.12	93.57	**95.04**
9	Sidewalk	**67.56**	62.24	53.03	53.27	52.81	62.40	65.22	53.75	67.18	58.71	60.18
10	Yellow Curb	80.12	71.41	73.68	81.10	43.87	74.79	**86.01**	78.96	74.60	75.03	78.22
11	Cloth Panels	95.34	94.34	94.82	95.54	**97.23**	92.49	95.26	92.73	95.30	92.17	96.83
	OA(%)	82.50	84.8	77.63	79.56	80.19	83.13	83.54	80.39	83.11	81.23	**86.08**
	AA(%)	82.72	82.87	77.98	80.39	79.95	82.33	83.62	79.21	82.09	80.75	**84.37**
	Kappa(%)	77.48	80.39	71.63	73.91	75.06	78.29	78.78	74.87	78.23	76.00	**81.93**

3) Results on Trento Dataset: Table III summarizes classification results on the Trento dataset. Within the scope of this comparative study, HFMNet attains the highest scores on all three evaluation metrics, demonstrating its balanced capacity to maintain consistently high accuracy across land-cover categories. Among all the categories, the model achieves optimal performance in one-third of them, which further validates the strong generalization ability of its feature extractor. Nevertheless, "Roads" classification accuracy is marginally lower due to spectral heterogeneity in road surfaces. Shadows

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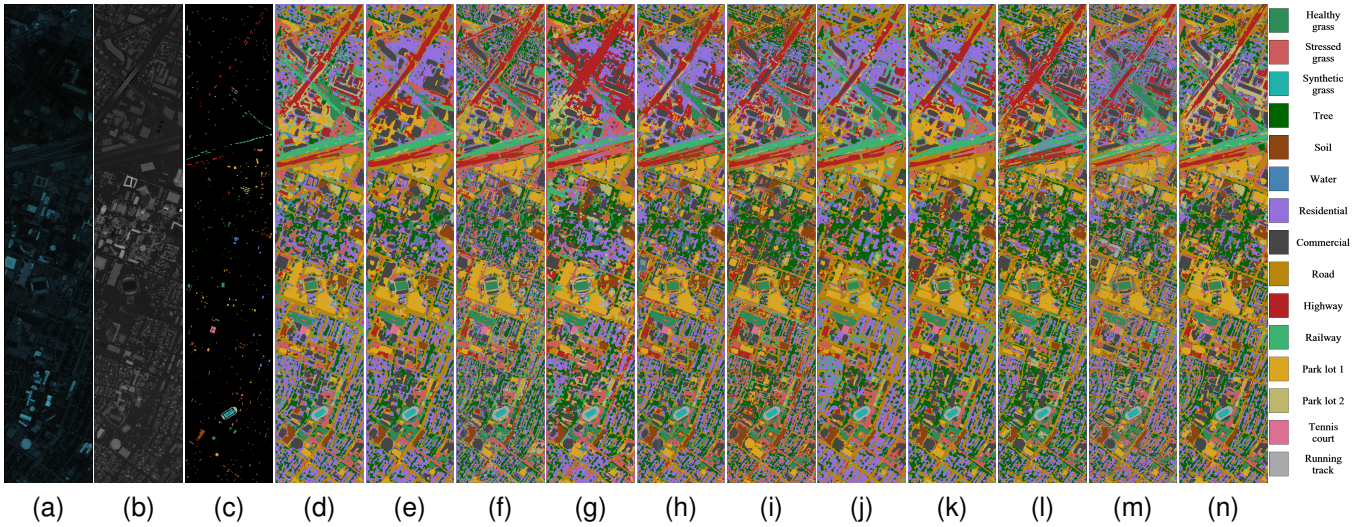


Fig. 3: Classification maps for all pixels in the Houston2013 dataset using different methods. (a) Pseudo-color composite image based on bands 9, 19, and 29 for HSI. (b) Grayscale image for LiDAR. (c) Ground-truth map. (d) Coupled-CNNs. (e) GLT-Net. (f) AMSSE-Net. (g) Fusion_HCT. (h) DSHFNet. (i) HFSNet. (j) MICF-Net. (k) MGMNet. (l) DCU-Net. (m) MSAF. (n) HFMNet.

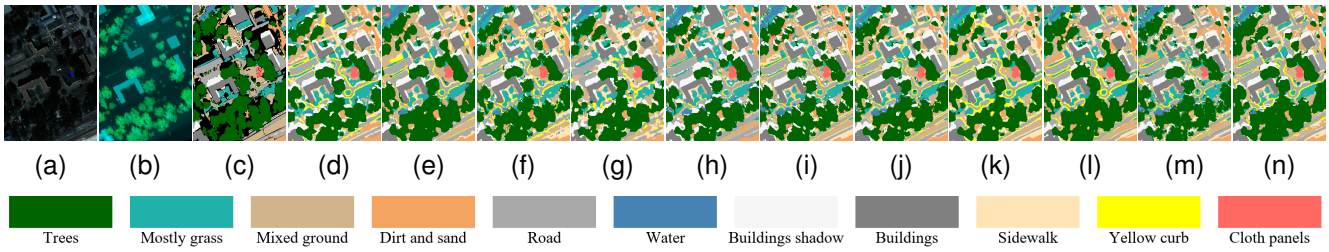


Fig. 4: Classification maps for all pixels in the MUUFL dataset using different methods. (a) Pseudo-color composite image based on bands 9,19, and 29 for HSI. (b) Grayscale image for LiDAR. (c) Ground-truth map. (d) Coupled-CNNs. (e) GLT-Net. (f) AMSSE-Net. (g) Fusion_HCT. (h) DSHFNet. (i) HFSNet. (j) MICF-Net. (k) MGMNet. (l) DCU-Net. (m) MSAF. (n) HFMNet.

TABLE III: Classification performance of different methods on Trento dataset (the best results are shown in **bold** and the second best ones are underlined)

No.	Class	Coupled-CNNs	GLT-Net	AMSSE-Net	Fusion_HCT	DSHFNet	HFSNet	MICF-Net	MGMNet	DCU-Net	MSAF	HFMNet
1	Apple Trees	98.92	99.12	98.82	98.64	**99.65**	99.38	98.93	99.09	99.15	98.87	99.03
2	Buildings	96.47	97.62	96.52	94.44	**99.23**	98.99	95.62	98.01	96.93	97.48	97.98
3	Ground	98.00	98.52	96.23	98.02	98.30	97.60	98.26	98.24	98.47	96.10	**98.58**
4	Woods	99.93	**100.00**	99.99	**100.00**	99.78	**100.00**	99.93	99.92	99.99	99.65	**100.00**
5	Vineyard	**100.00**	**100.00**	99.71	98.08	99.98	99.80	99.94	99.98	99.99	93.34	99.95
6	Roads	97.02	96.30	97.22	93.36	94.78	95.88	97.47	96.92	**97.75**	93.94	97.27
OA(%)		99.15	99.24	99.06	97.89	99.23	98.78	99.10	99.31	99.33	96.49	**99.35**
AA(%)		98.39	98.59	98.08	97.09	98.62	97.91	98.36	98.69	98.71	96.56	**98.80**
Kappa(%)		98.87	98.99	98.74	97.19	98.98	98.37	98.80	99.08	99.10	95.34	**99.13**

and material variations create confounding effects that obscure edge details. Fig. 5 presents pixel-level classification maps from different methods.

E. Ablation Studies

The proposed architecture comprises two core sub-modules, namely HFME and ML-DAGF. To assess the individual contributions of these modules, we perform ablation experiments in which each sub-module is successively replaced with a standard convolutional layer. This replacement preserves the original network connectivity while ensuring valid classification outputs. The resulting ablated variants are denoted as "w/o HFME" and "w/o ML-DAGF", respectively, and their results on the MUUFL dataset are reported in Table IV. HFME yields a consistent improvement, which is attributed to its capability to capture global image information and enhance

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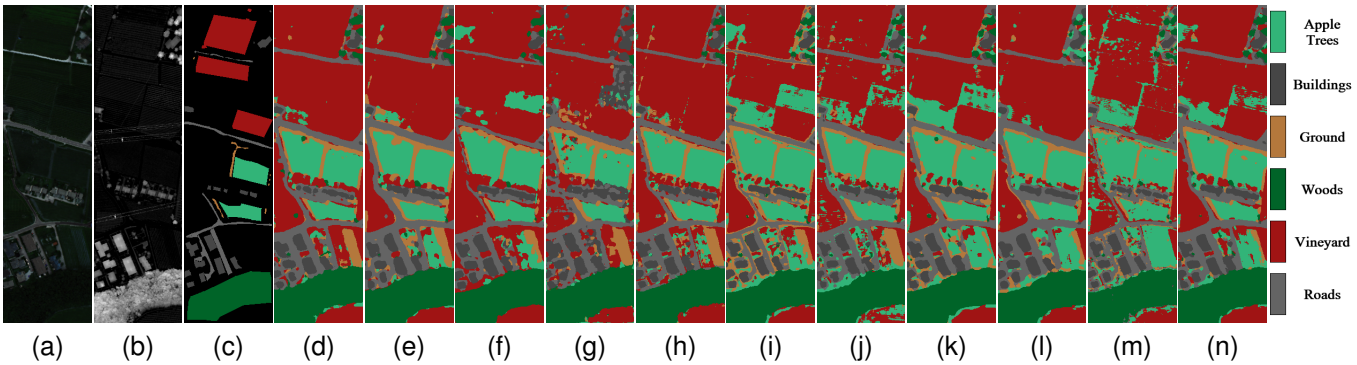


Fig. 5: Classification maps for all pixels in the Trento dataset using different methods. (a) Pseudo-color composite image based on bands 9, 19, and 29 for HSI. (b) Grayscale image for LiDAR. (c) Ground-truth map. (d) Coupled-CNNs. (e) GLT-Net. (f) AMSSE-Net. (g) Fusion_HCT. (h) DSHFNet. (i) HFSNet. (j) MICF-Net. (k) MGMNet. (l) DCU-Net. (m) MSAF. (n) HFMNet.

TABLE IV: Ablation Study of The HFMNet on The MUUFL Dataset

Method	MUUFL		
	OA(%)	AA(%)	Kappa(%)
Baseline	82.29	80.74	77.14
w/o HFME	85.43	82.99	81.09
w/o ML-DAGF	82.65	81.64	77.62
HFMNet	**86.08**	**84.37**	**81.93**

feature discriminability. ML-DAGF also exhibits the most pronounced performance impact. This indicates that ML-DAGF effectively exploits spatial and channel-wise complementary information and aggregates it in a coordinated manner. The joint integration of HFME and ML-DAGF further enhances the overall model performance, reflecting the complementary roles of global feature extraction and multi-level attention-based fusion. Specifically, ML-DAGF captures complementary multimodal semantics via attention mechanisms and enables smooth feature transitions through multi-level aggregation. Consequently, this design alleviates information loss and insufficient fusion during multimodal feature integration.

V. Conclusion

This paper proposes HFMNet for multimodal remote sensing image classification. The network effectively extracts global and local features from multimodal data and selectively fuses multi-level complementary information along both spatial and channel dimensions. First, convolutional neural networks are employed to extract spatial and channel-wise features from HSI and LiDAR data. Next, Spatial Mamba models global-local dependencies in multimodal data, while a Fourier-based Frequency-Enhanced module captures high-frequency features. Together, these modules extract modality-specific information from complementary perspectives. Finally, multi-level dual-attention-guided fusion integrates multimodal features by jointly leveraging spatial and channel attention mechanisms. Following integration of complementary multimodal information, the fused features are passed to a fully-connected classifier for land-cover prediction. The network's effectiveness is validated through extensive experiments on three datasets.

Although the proposed model achieves competitive performance, there remains room for further improvement. Future work may focus on incorporating contour-aware representations or more fine-grained texture extraction modules to better characterize complex land-cover categories.

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